

Tomohiko Nakamura

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Research Interests

Signal-processing-inspired deep learning, audio and music signal processing, and machine learning

Job

Senior Researcher <i>Intelligent Media Processing Research Team, Artificial Intelligence Research Center, National Institute of Advanced Industrial Science and Technology (AIST)</i>	Apr. 2023–Present
Visiting Researcher <i>Graduate School of Information Science and Technology, The University of Tokyo</i>	Nov. 2025–Present
Project Research Associate <i>Graduate School of Information Science and Technology, The University of Tokyo</i>	Sep. 2019–Mar. 2023
Researcher <i>Intelligent Systems Laboratory, SECOM</i>	Apr. 2016–Aug. 2019
Research Fellow (DC2) <i>Japan Society for the Promotion of Science (JSPS)</i>	Apr. 2015–Mar. 2016

Education

Ph.D. degree in Information Science and Technology <i>Graduate School of Information Science and Technology, The University of Tokyo</i>	Mar. 2016
Master's degree in Information Science and Technology <i>Graduate School of Information Science and Technology, The University of Tokyo</i>	Mar. 2013
Bachelor's degree in Engineering <i>Faculty of Engineering, The University of Tokyo</i>	Mar. 2011

Teaching

Advanced Signal Processing <i>Graduate School of Information Science and Technology, The University of Tokyo</i>	June 2020, 2022, 2024, and 2026
Applied Gaussian Process and Machine Learning <i>Graduate School of Information Science and Technology, The University of Tokyo</i>	Dec. 2021
Keisu Student Experiment <i>Department of Mathematical Engineering and Information Physics, The University of Tokyo</i>	Apr. 2020–Mar. 2023

Languages

Japanese (native), English (basic)

Competitive Funds

JST PRESTO (Fundamental Innovation for Real-World Intelligent Systems) <i>Variable time-resolution audio processing framework adaptive to acoustic scenes</i>	Oct. 2025–Mar. 2029
JSPS KAKENHI Grant-in-Aid for Scientific Research (B) <i>Development of deep-layered analysis-by-synthesis techniques for acoustic scene analysis with human intervention</i>	Apr. 2023–Mar. 2027
JST ACT-X (Frontier of Mathematics and Information Science)	

+ 4 funds as representative, 4 funds as co-researcher, and 2 funds as research participant.

Publications

Journal Papers.....

- [1] Yoshiaki Bando, Tomohiko Nakamura, Satoru Fukayama, and Shinji Watanabe, "Online frontend system for multi-talker DSR using neural blind source separation and diarization," *IEEE Transactions on Audio, Speech and Language Processing*, vol. 34, pp. 3698–3713, July 2026.
- [2] Ren Uchida, Kohei Yatabe, and Tomohiko Nakamura, "Encoder-masking-decoder networks using orthogonal convolutional layer as invertible linear encoder," *Acoustical Science and Technology*, vol. advpub, no. e26.10, May 2026.
- [3] Kanami Imamura, Tomohiko Nakamura, Norihiro Takamune, Kohei Yatabe, and Hiroshi Saruwatari, "Stride conversion algorithms for convolutional layers and its application to sampling-frequency-independent deep neural networks," *Signal Processing*, vol. 242, Nov. 2025.
- [4] Yuki Ito, Tomohiko Nakamura, Shoichi Koyama, Shuichi Sakamoto, and Hiroshi Saruwatari, "Spatial upsampling of head-related transfer function using neural network conditioned on source position and frequency," *IEEE Open Journal of Signal Processing*, vol. 6, pp. 1109–1123, Sept. 2025.
- [5] +11 papers

Peer-Reviewed International Conferences and Workshops.....

- [1] Kengo Takemoto, Tomohiko Nakamura, and Hiroshi Saruwatari, "Differentiable digital signal processing mixture model-based diffusion for synthesis parameter estimation from harmonic sound mixtures," in *Proceedings of Asia Pacific Signal and Information Processing Association Annual Summit and Conference*, Nov. 2026.
- [2] Yuma Narahata, Tomohiko Nakamura, Yuki Saito, and Hiroshi Saruwatari, "Lead vocal separation from vocal ensemble mixtures using phoneme alignment," in *Proceedings of Asia Pacific Signal and Information Processing Association Annual Summit and Conference*, Nov. 2026.
- [3] Tomohiko Nakamura, Wataru Nakata, Kanami Imamura, and Yuki Saito, "Neural audio codec with adjustable token temporal resolution using sampling-frequency-independent convolutional layers," in *Proceedings of International Workshop on Acoustic Signal Enhancement*, Sept. 2026.
- [4] +47 papers

Invited Talks.....

- [1] Tomohiko Nakamura, "Trends and prospects for audio source separation using deep learning," *Meeting on Technical Committee on Engineering Acoustics, IEICE*, vol. 124, no. 389, EA2024-93, p. 104, Mar. 2025. (in Japanese)
- [2] Daichi Kitamura and Tomohiko Nakamura, "Fundamentals and applications of audio source separation — A guide to becoming an expert," *2023 Otagaku Symposium*, vol. 2023-MUS-137 no. 35, Jun. 2023. (in Japanese)
- [3] Tomohiko Nakamura, "Signal-processing-inspired deep learning," *IEEE NZ Signal Processing/Information Theory Joint Chapter, co-hosted by the Acoustics Research Centre, University of Auckland*, Dec. 2022.
- [4] +1 invited presentations

Overview Papers.....

1. Shoichi Koyama, Juliano Ribeiro, Tomohiko Nakamura, Natsuki Ueno, and Mirco Pezzoli, "Physics-informed machine learning for sound field estimation," *Special Issue on Model-Based and Data-Driven Audio Signal Processing, IEEE Signal Processing Magazine*, vol. 41, pp. 60–71, Nov. 2024.

Patents.....

- [1] Tomohiko Nakamura, "Object recognition device, method, and program," Japan Patent JP7349288, 13-Sept-2023.
- [2] +10 patents

Awards

1. The Awaya Kiyoshi Research Award, ASJ, Mar. 2024.
2. The Itakura Prize Innovative Young Researcher Award, ASJ, Mar. 2022.
3. +12 awards